

The Mirror Branch Braid Atlas: A Parameter-Free Dark Sector from the Universal Generative Principle

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Abstract

The Universal Generative Principle (UGP) derives the Standard Model spectrum from a parameter-free arithmetic sieve at ridge $n = 10$. The prime-lock mechanism that selects the SM lepton seed $(1, 73, 823)$ simultaneously forces a mirror-branch seed $(1, 73, 2137)$, predicting a complete dark sector with no free parameters.

We derive the Mirror Branch Braid Atlas—the dark sector particle classification table. The electric charge $Q = 0$ for all dark particles is derived from Braid Atlas topology, not assumed: the mirror $\mathbb{Z}_2$ involution is an internal GTE arithmetic symmetry, not an SM gauge transformation. The dark sector gauge group is $SU(3)_{\text{dark}}$ only—no dark $SU(2)_L$ exists. Three generations of free dark leptons at 0.54 MeV, 24.5 MeV, and 3.60 GeV coexist with confined dark quarks near $\Lambda_{\text{dark}} \sim 200$ MeV.

We establish a new structural connection: $c(W) = b_2(\text{RHN}) = q_1$ in both branches—the first cross-sector structural unification in UGP Physics. The core results are machine-certified in Lean 4 (`MirrorWindingNumber.lean`, `EWBosonRHNConnection.lean`, `DarkBraidAtlas.lean`; zero sorry, zero custom axioms).

The dark neutrino mass-squared ratio $R_{\text{dark}} = 0.208$ is $7\times$ larger than the SM ratio (conditional on the structural gap theorem). The FIMP mechanism overproduces dark matter by 4.84×10^6 at $\lambda_s \sim 10^{-6}$; the leading resolution is asymmetric dark matter with $\eta_\chi \approx (2/7)\eta_{B+L,\text{pre}}$, where $2/7 = (\mathbb{Z}_7 \text{ charge of dark baryon})/(\mathbb{Z}_7 \text{ group order})$. The most accessible prediction is GTE-P7 at 211.9 MeV—a Tier 1 target at Belle II. The baryogenesis derivation and dark confinement scale are identified as primary open problems.

Contents

1	Introduction and Mirror Branch Structure	2
1.1	The UGP Dark Sector Prediction	2
1.2	Contributions of This Paper	3

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1.3	Comparison with Existing Dark Sector Models	3
1.4	Lean Certificate Summary	4
1.5	Paper Structure	4
2	Mirror Branch Braid Atlas: Dark Sector Classification	5
2.1	Mirror Branch Structure	5
2.2	Braid Topology of Mirror Particles	5
2.3	Dark Singlet Leptons: Three Generations, $Q = 0$ Derived	5
2.4	Dark Quarks: $Q = 0$, Dark $SU(3)$ Colour, Confinement	6
2.5	Two-Component Dark Sector	7
2.6	The Zero-Parameter Distinction	7
3	EW Boson–RHN Structural Connection	8
3.1	The Connection in the SM Branch	8
3.2	Extension to the Mirror Branch	8
3.3	Physical Interpretation	9
3.4	No Dark Electroweak Symmetry	9
4	Dark Sector Neutrino Mass Structure	9
4.1	Mirror RHN b -Values	9
4.2	Dark Neutrino Mass Hierarchy	10
4.3	Physical Discussion	10
5	Experimental Predictions and Detection Prospects	11
5.1	Prediction Table	11
5.2	Dark Hadron Subsector and the Confinement Scale	11
5.3	Current Experimental Status and the Dark Photon Non-Prediction	12
5.4	The Two-Peak Signature	12
5.5	Relic Density: FIMP Overproduction and the Asymmetric Dark Matter Mechanism	13
6	Conclusions	14
	Appendices	15
A	Lean Certificate Inventory	15
B	Dark Sector Mass Formula	16

1 Introduction and Mirror Branch Structure

1.1 The UGP Dark Sector Prediction

The Universal Generative Principle (UGP) is a parameter-free framework that derives the Standard Model particle spectrum from an arithmetic sieve operating at ridge $n = 10$ [8,13]. The sieve selects a unique lepton seed $(1, 73, 823)$ via three simultaneous constraints: prime-lock (both c_1 and its mirror must be prime), mirror-duality (a $b_2 \leftrightarrow q_2$ exchange symmetry of the cascade), and minimal description length (MDL) minimality. The resulting Braid Atlas assigns particle types to GTE orbits and derives Standard Model quantum numbers from arithmetic structure [9].

A central result of this paper is that the same prime-lock mechanism that selects the SM seed simultaneously *requires* a mirror branch—a second arithmetic solution at ridge $n = 10$ with $b_2 \leftrightarrow q_2$ exchanged: seed $(1, 73, 2137)$. This mirror branch is not optional; it is forced by the same sieve that forces the SM. The two seeds form an arithmetic pair: neither can be accepted without the other.

The mirror branch predicts a complete dark sector with specific, parameter-free particle masses and quantum numbers. Unlike conventional dark sector models where masses are free parameters, the UGP dark sector masses emerge from the same arithmetic that determines the SM masses—with zero additional freedom.

1.2 Contributions of This Paper

This paper establishes the following results:

1. The Mirror Braid Atlas: particle classification for all dark sector states, with quantum number assignments derived (not assumed) from GTE arithmetic.
2. $Q_{\text{EM}} = 0$ for all dark sector particles, derived from the Braid Atlas winding number theorem (not assumed), machine-certified in Lean 4 with zero custom axioms.
3. Two dark subsectors: free dark singlet leptons (3 generations, FIMP sector) and confined dark quarks forming dark hadrons at scale Λ_{dark} .
4. The EW Boson–RHN structural connection: $c(W) = b_2(\text{RHN}) = q_1$ in both branches (first cross-sector structural unification; Lean-certified).
5. The dark neutrino mass-squared ratio $R_{\text{dark}} = 0.208$, compressed $7\times$ relative to the SM (conditional on the structural gap theorem).
6. The relic density overproduction factor 4.84×10^6 at $\lambda_s \sim 10^{-6}$, and the leading asymmetric dark matter resolution with $\eta_\chi \approx (2/7)\eta_{B+L}$.
7. Experimental predictions and detection prospects, including the two-peak signature at 0.54 MeV and 24.5 MeV.

1.3 Comparison with Existing Dark Sector Models

The UGP dark sector differs from all existing dark sector proposals in several decisive respects. Table 1 summarizes the key distinctions.

Table 1: Comparison of the UGP mirror branch with representative dark sector models. “ $Q = 0$ ” refers to the electric charge of the lightest dark state.

Model	$Q = 0$ mechanism	Mass spectrum	Dark gauge group
WIMP	Assumed	Free (GeV–TeV)	Free
Axion	By construction	Free (μeV –meV)	$U(1)_{\text{PQ}}$
Sterile neutrino	Assumed	Free (keV–GeV)	None
Dark photon	Assumed	Free, ε free	Dark $U(1)_D$
Mirror matter [3]	$Q = -1$ (excluded)	Free	Copy of SM
UGP mirror branch	Derived from braid topology	Fixed: 0.54, 24.5, 3600 MeV	$SU(3)_{\text{dark}}$ only

The decisive distinction is that in UGP, $Q = 0$ is a *theorem* derived from the braid winding number, not an assumption. Mirror matter models (Foot, Lew, Volkas [3]) assign $Q = -1$ to the

dark electron by analogy with the SM; this is excluded by BBN and CMB constraints because mirror-charged particles would have thermalized with mirror photons and left a detectable radiation imprint. The UGP mirror branch avoids this because the $b_2 \leftrightarrow q_2$ swap maps GTE arithmetic invariants, not SM gauge charges: the dark particles are SM gauge singlets, giving $Q = 0$ as a structural consequence.

1.4 Lean Certificate Summary

All central structural claims have Lean 4 machine-checked proofs in the UGP Lean repository [14]. Table 2 summarizes the key certificates.

Table 2: Lean 4 certificates for central claims. All files are in the `UgpLean/BraidAtlas/` or `GTE/` directories of the UGP Lean repository. “Zero sorry” means the proof compiles with no admitted gaps; “zero axioms” means no custom axioms beyond Lean/Mathlib foundations.

Lean file	Key theorem(s)	Status
<code>MirrorWindingNumber.lean</code>	$W_g = 0, Q = 0$ for dark leptons (all gens); orbit distinguishability	Zero axioms
<code>DarkQuarkCharge.lean</code>	<code>dark_quark_charge_zero</code> (all gens)	Zero sorry, zero axioms
<code>DarkBraidAtlas.lean</code>	$b'_1 = 5, b'_2 = 29, q'_1 = 29$; $SU(2)_L$ singlet structure	Zero sorry (inherits definitions from P17 canonical modules)
<code>EWBosonRHNConnection.lean</code>	$c(W) = b_2(\text{RHN}) = q_1$ (both branches)	Zero sorry, zero axioms
<code>RHNGapTheorem.lean</code>	$19 = 11 + 8, 37 = 29 + 8$ (arithmetic certificate)	Zero sorry
<code>GTBGenerationPrimes.lean</code>	All 6 generation c_1 values prime; 3-generation structure	Zero sorry

The Lean certification distinguishes this work from all prior dark sector proposals: the structural claims are not merely argued but machine-verified against formal definitions. An open Lean item is the structural gap theorem deriving $b_3 = q_2 - b_{1,\text{RHN}}$ from GTE cascade rules; until this is proved, the dark neutrino hierarchy is conditional.

1.5 Paper Structure

Section 2 derives the Mirror Braid Atlas and establishes the dark particle spectrum, including $Q = 0$ for dark leptons and dark quarks. Section 3 proves the EW Boson–RHN structural connection and its mirror-branch extension. Section 4 derives the dark neutrino mass hierarchy. Section 5 presents experimental predictions: the two-peak MeV signature, dark confinement structure, relic density status, and detection prospects. Section 6 concludes. Appendix A provides the full Lean certificate inventory.

2 Mirror Branch Braid Atlas: Dark Sector Classification

2.1 Mirror Branch Structure

The GTE arithmetic sieve at ridge $n = 10$ has exactly two outputs satisfying all prime-lock constraints: the SM sector and the mirror sector. Both arise from the same prime-lock mechanism; neither can be accepted without the other.

The two branches are distinguished by the swap $b_2 \leftrightarrow q_2$, which exchanges two outputs of the same prime-lock computation. The arithmetic parameters are:

Parameter	SM branch	Mirror branch
b_2	42	24
q_2	24	42
c_1	823	2137
q_1	11	29

Both $c_1 = 823$ and $c'_1 = 2137$ are prime (Lean-certified). The mirror seed satisfies $c'_1 = b'_2 \cdot q'_2 + 20 = 24 \times 89 + 1 = 2137$ with residue $2137 \bmod 73 = 20$, matching the SM residue class. The pair is unique: `prime_lock_unique_survivor` in the UGP Lean 4 formalization library [14] certifies that no other pair at ridge $n = 10$ satisfies all constraints simultaneously.

The swap $b_2 \leftrightarrow q_2$ is an arithmetic symmetry of the GTE cascade integers. It is *not* an SM gauge transformation: mirror-branch particles carry no SM gauge charges [9]. This is the physical origin of the dark sector’s electrical neutrality.

The branch-shift invariant is $\Delta q \equiv q_2 - q_1 = 24 - 11 = 13 = q'_2 - q'_1 = 42 - 29$. This spacing is preserved across branches and provides an internal consistency check on the arithmetic structure.

2.2 Braid Topology of Mirror Particles

Mirror-branch particles occupy the same braid orbit structure as their SM analogues. Dark leptons correspond to the single-strand ($a_1 = 1$) orbit class, analogous to SM charged leptons and neutrinos. Dark quarks are single-strand under the dark $SU(3)$, though SM $SU(3)_c$ acts trivially on them (dark quarks carry dark color, not SM color).

The mirror orbit ($b'_2 = 24, q'_2 = 42, c'_1 = 2137$) is arithmetically distinct from the SM lepton orbit ($b_2 = 42, q_2 = 24, c_1 = 823$): same residue class $m_1 = 20$ (same GTE orbit family), different representatives ($c'_1 = 2137 > 823 = c_1$), and different first quotients ($q'_1 = 29 \neq 11 = q_1$). This is certified by `MirrorWindingNumber.lean` (zero axioms): the orbit distinguishability theorems `mirror_c1_ne_lepton_c1` and `mirror_q1_ne_canonical_sm_q1` confirm the arithmetic separation.

2.3 Dark Singlet Leptons: Three Generations, $Q = 0$ Derived

The mirror branch generates three generations of dark singlet leptons. Their quantum numbers and masses are given in Table 3.

Table 3: Dark singlet lepton spectrum. Masses are parameter-free predictions of the GTE cascade arithmetic applied to the mirror branch (not Lean-certified at the mass level; see text). The charge $Q_{\text{EM}} = 0$ is a Lean-certified theorem (zero axioms).

Generation	Label	Mass	Q_{EM}	Primary search venue
G1	χ_1	0.5406 MeV	0	Sub-MeV dark matter (Higgs portal)
G2	χ_2	24.47 MeV	0	MeV-scale dark matter (Higgs portal)
G3	χ_3	3604.68 MeV	0	LHC Higgs invisible / mono-jet (λ_s suppressed)

Derivation of $Q = 0$ (not assumed). The mirror orbit has $Y_{\text{mirror}} = 0$ and $T_{3,\text{mirror}} = 0$ (no SM hypercharge, no SM weak isospin) by the universal mirror-duality statement of P17 [9]: the $b_2 \leftrightarrow q_2$ swap is an internal GTE arithmetic operation, not an SM gauge transformation, so mirror-branch particles carry no SM gauge quantum numbers. The standard Gell-Mann–Nishijima formula $Q = T_3 + Y/2$ then gives $Q = 0 + 0 = 0$. In the UGP GTE convention, the *winding number* $W_g \equiv N_c \times Q$ (an integer multiple of the charge by the colour rank $N_c = 3$) satisfies:

$$W_g = N_c \times (T_3 + Y/2) = 3 \times 0 = 0 \implies Q = W_g/N_c = 0. \quad (1)$$

This is a theorem from definitions, not an axiom. The Lean 4 proof is in `MirrorWindingNumber.lean` (zero custom axioms): the theorem `mirror_lepton_q_em_zero` proves $Q = 0$ for all three generations simultaneously via `dark_lepton_q_em_zero_all_gens`.

Why this differs from mirror matter models. Mirror matter proposals [3] assign $Q = -1$ to the dark electron because they map the SM electron—including its SM gauge charges—to a mirror partner. In the UGP mirror branch, the $b_2 \leftrightarrow q_2$ swap operates on GTE arithmetic labels, not on SM gauge representations. The dark particles are SM gauge singlets from the outset. There is no mirror electric charge, no mirror photon, and no mirror electromagnetic coupling. The charge $Q = 0$ follows from $Y_{\text{mirror}} = 0$ as a structural consequence of the GTE prime-lock.

Dark lepton mass hierarchy. The three-generation mass hierarchy satisfies:

$$\frac{G2}{G1} = \frac{24.47}{0.5406} \approx 45.3 \quad \text{vs. SM} \quad \frac{m_\mu}{m_e} \approx 206.8, \quad (2)$$

$$\frac{G3}{G2} = \frac{3604.68}{24.47} \approx 147.3 \quad \text{vs. SM} \quad \frac{m_\tau}{m_\mu} \approx 16.8. \quad (3)$$

The G2/G1 ratio is compressed by a factor of approximately 4.6 relative to the SM charged lepton hierarchy. This compression is not tuned; it is determined by the mirror branch GTE values ($b'_2 = 24$ vs. SM $b_2 = 42$). The G3/G2 ratio is *enhanced* relative to SM τ/μ , producing an asymmetric overall hierarchy that is distinctive. Both ratios are falsifiable predictions.

2.4 Dark Quarks: $Q = 0$, Dark SU(3) Colour, Confinement

Dark quarks carry $Q_{\text{EM}} = 0$. This follows from the same universal $Y_{\text{mirror}} = 0$ argument as for dark leptons. The key distinction is that dark quarks *do* carry a dark SU(3) colour charge (distinct from SM SU(3)_c), but this internal dark charge does not contribute to the Gell-Mann–Nishijima formula, which is defined with respect to SM quantum numbers.

The Lean 4 proof is in `DarkQuarkCharge.lean` (zero sorry, zero axioms), which contains three key theorems: (i) `dark_quark_charge_zero`, which proves $W_g^{\text{mirror}} = 0$ from the mirror winding

number; (ii) `dark_quark_charge_zero_all_gens`, which extends the result to all generations; and (iii) a composite machine-checked certificate that the quark charge result holds across all mirror-branch generations simultaneously. The proof is independent of the lepton case—it establishes $Q = 0$ for the dark quark sector separately.

Table 4 contrasts SM quarks with their dark counterparts.

Table 4: SM quarks vs. dark quarks: quantum number comparison. Dark quarks carry dark SU(3) colour (not SM colour) and have $Q_{\text{EM}} = 0$ (Lean-certified).

Particle	W_g	Q_{EM}	Colour
SM up quark	+2	+2/3	SM SU(3) _c
SM down quark	−1	−1/3	SM SU(3) _c
Dark up quark G1	0	0	Dark SU(3)
Dark down quark G1	0	0	Dark SU(3)

The dark quark G1 masses are 0.57 MeV (dark up) and 17.30 MeV (dark down), derived from the mirror-branch Permutation Principle (preliminary, pending verification from the locked GTE map). Dark G2 and G3 quark masses are not yet computable from current GTE results. Since dark quarks carry dark SU(3) colour, they confine at the dark confinement scale Λ_{dark} into dark hadrons (dark pions, dark protons); see Section 5.2.

2.5 Two-Component Dark Sector

The mirror branch produces a two-component dark sector:

Component 1: Free dark singlet leptons (FIMP sector). Dark singlet leptons (χ_1, χ_2, χ_3) are electrically neutral (Lean-certified), dark SU(3) singlets (they are not quarks and do not confine), and produced via freeze-in (FIMP mechanism) through the Higgs portal. The FIMP mechanism—introduced in [4, 6]—is specifically designed for very small couplings ($\lambda_s \sim 10^{-6}$, two-loop suppressed), which is precisely the UGP portal coupling.

Component 2: Confined dark hadrons (dark strong sector). Dark quarks carry dark SU(3) colour and confine at Λ_{dark} into dark hadrons. The confinement structure is discussed in Section 5.2.

2.6 The Zero-Parameter Distinction

Every existing dark sector model introduces at least one free parameter: WIMP mass, axion decay constant, sterile neutrino mixing angle, dark photon kinetic mixing. The UGP mirror branch has zero free parameters (Table 5).

Table 5: Free parameter count in representative dark sector models vs. UGP.

Model	Free params	$Q = 0$ status	DM mass range
WIMP	≥ 2 (mass + coupling)	Assumed	GeV–TeV
Axion	≥ 2 (mass + f_a)	By construction	μeV –meV
Sterile ν	≥ 2 (mass + mixing)	Assumed	keV–GeV
Dark photon	≥ 2 (mass + ε)	Assumed	MeV–GeV
Mirror matter [3]	≥ 1 (scale)	$Q = -1$ (excluded)	Broad
UGP mirror branch	0	Derived (Lean 4, zero axioms)	0.54, 24.5, 3600 MeV

The UGP dark sector is not a model but a prediction: the arithmetic sieve that selects the SM necessarily produces this dark sector with the same number of free parameters—zero.

3 EW Boson–RHN Structural Connection

3.1 The Connection in the SM Branch

Computing separately from the GTE cascade at ridge $n = 10$ the c -value of the W boson and the b_2 parameter of the right-handed neutrino sector—two outputs from different stages of the cascade that were not previously recognized as related—we establish:

$$c(W) = b_2(\text{RHN}) = q_1 = 11. \quad (4)$$

The common value is $q_1 = 11$, the first GTE quotient at ridge $n = 10$. The identity $c(W) = q_1$ follows from the W boson c -value computation in the SM branch [9]; the identity $b_2(\text{RHN}) = q_1$ follows from the RHN cascade step [12]. That both equal the same q_1 is structural: both are outputs of the GTE prime-lock at the same ridge, indexed by the first quotient.

The Lean 4 certificate is `EWBosonRHNConnection.lean` (zero sorry, zero axioms—the strongest possible certification). The composite theorem `ew_rhn_connection_both_branches` states:

$$c_W = b2_RHN \wedge c_W = q1$$

This connection was not stated or proved in the preceding UGP papers. P29 is the first to identify and formally certify this structural link.

3.2 Extension to the Mirror Branch

The connection extends exactly to the mirror branch with mirrored values. Since $q'_1 = 29$ for the mirror branch (see Section 2.1), and $b'_2(\text{RHN}) = 29$ (Lean-certified via `gteQuotient` in `DarkBraidAtlas.lean`), we have:

$$q'_1(\text{mirror}) = b'_2(\text{RHN}) = 29. \quad (5)$$

The branch shift is preserved: $q'_1(\text{mirror}) - q_1(\text{SM}) = 29 - 11 = 18$. The branch-invariant spacing $\Delta q \equiv q_2 - q_1 = 13$ is the same in both branches (SM: $24 - 11 = 13$; mirror: $42 - 29 = 13$).

It is important to note that $q'_1 = 29$ in the mirror branch is a GTE arithmetic invariant—it is *not* the c -value of a dark W' boson. The dark sector has no electroweak symmetry (Section 3.4), so no dark W' exists. The value 29 labels the arithmetic root of the mirror branch, and its coincidence with $b'_2(\text{RHN})$ is the structural connection being reported.

3.3 Physical Interpretation

The EW gauge sector (W boson) and the neutrino mass sector (right-handed neutrino) share the same arithmetic root q_1 . This is the first cross-sector structural unification in the UGP Physics framework.

In ordinary quantum field theory, there is no expectation that gauge boson parameters and fermion mass parameters share a common value—they enter from independent Lagrangian terms and are treated as unrelated free parameters. In the GTE framework, both W boson c -values and RHN b -values are outputs of the same cascade at the same ridge. Their shared arithmetic root q_1 is a consequence of GTE architecture.

Conjecture 3.1 (GTE Seesaw). *The identity $c(W) = b_2(\text{RHN}) = q_1$ may be a GTE-level hint about the origin of the type-I seesaw mechanism. In the type-I seesaw, the light neutrino mass $m_\nu \sim v^2/M_R$ involves the electroweak VEV v (set by the W boson mass) and the heavy RHN mass M_R . If both m_W and M_R share the arithmetic root q_1 , the seesaw mass relation may have a natural GTE explanation. This conjecture is not proved in P29 and is flagged as an open problem for future work.*

3.4 No Dark Electroweak Symmetry

A corollary of the $Q = 0$ derivation is that the dark sector has *no* electroweak symmetry. Since $T_{3,\text{mirror}} = Y_{\text{mirror}} = 0$ for all mirror-branch particles (Lean-certified in `DarkBraidAtlas.lean`), all dark particles are $\text{SU}(2)_L$ singlets. There is no dark $\text{SU}(2)_L$, no dark $\text{U}(1)_Y$, and no dark Higgs mechanism. The dark sector gauge group is:

$$G_{\text{dark}} = \text{SU}(3)_{\text{dark}} \quad (\text{only}). \quad (6)$$

This rules out a dark W' , dark Z' , or dark Higgs boson from an electroweak mechanism. It also rules out kinetic mixing between a dark photon and the SM hypercharge boson: since no dark $\text{U}(1)_D$ exists, the kinetic mixing parameter $\varepsilon = 0$ exactly by structure (not by fine-tuning). This is a falsifiable structural prediction: a discovery of a dark photon signal would falsify the UGP dark sector at the highest confidence level.

4 Dark Sector Neutrino Mass Structure

4.1 Mirror RHN b -Values

The mirror branch GTE cascade produces three right-handed neutrino b -values, which govern the dark neutrino mass hierarchy via the same mass formula established in P21 [12]. The values and their certification status are:

Parameter	Value	Certificate	Notes
$b'_1(\text{RHN})$	5	Branch-invariant	$b_1 = 2N_c - 1 = 5$; holds for both branches
$b'_2(\text{RHN})$	29	Lean-certified	<code>dark_b2_rhn_eq_29</code> in <code>DarkBraidAtlas.lean</code>
$b'_3(\text{RHN})$	37	Arithmetic cert	$37 = 29 + 8$; structural derivation open

The branch invariance of $b_1 = 5$ follows from the N_c formula: $b_1 = 2N_c - 1 = 5$ regardless of which branch is selected, because $N_c = 3$ is an SM structural constant shared by both branches.

The value $b'_3 = 37$ is arithmetically certified: `RHNGapTheorem.lean` (zero sorry) proves $37 = 29 + 8$, confirming that the gap formula $b_3 = b_2 + (N_c^2 - 1) = b_2 + 8$ holds arithmetically in both branches. However, the structural derivation—why $b_3 = q_2 - b_{1,\text{RHN}}$ from GTE cascade rules—is not yet proved. Until this structural derivation is complete, $b'_3 = 37$ is conditional and all results depending on it are subject to this caveat.

4.2 Dark Neutrino Mass Hierarchy

Given the mirror RHN b -values, the dark neutrino mass-squared ratio is:

$$R_{\text{dark}} = \frac{b_2'^{58/9} - b_1'^{58/9}}{b_3'^{58/9} - b_1'^{58/9}} = \frac{29^{58/9} - 5^{58/9}}{37^{58/9} - 5^{58/9}} = 0.2080. \quad (7)$$

This formula is the same as the one derived for the SM in P21 [12], applied to the mirror branch b -values. The exponent $58/9$ emerges from the GTE cascade arithmetic and is treated here as branch-invariant—the same structural formula that applies to the SM branch is applied to the mirror branch. This is an assumption, not a derived result: a first-principles derivation of the exponent for the mirror branch separately remains an open item. The SM ratio, $R_{\text{SM}} = 0.02936$, is Lean-certified in P21 and matches NuFIT 6.0 [2] to within 0.16σ .

Quantity	Value	Certificate	Notes
R_{SM}	0.02936	Lean-certified (P21)	Matches NuFIT 6.0 at 0.16σ
R_{dark}	0.2080	Conditional on $b'_3 = 37$	See caveat below
$R_{\text{dark}}/R_{\text{SM}}$	≈ 7.09	Conditional	Dark hierarchy $7\times$ larger

The dark neutrino hierarchy follows normal ordering ($b'_1 < b'_2 < b'_3$), the same ordering as the SM. The hierarchy is compressed at the G1/G2 step (where SM and dark differ most) and enhanced at the G2/G3 step.

Caveat: All claims about R_{dark} are conditional on the structural gap theorem deriving $b'_3 = q'_2 - b'_1$ from first-principles GTE cascade rules. The arithmetic certificate (`RHNGapTheorem.lean`) establishes $37 = 29 + 8$ but does not prove the GTE origin of this gap. Until that structural derivation is complete, $R_{\text{dark}} = 0.2080$ should be understood as a conditional prediction.

4.3 Physical Discussion

The compression factor $R_{\text{dark}}/R_{\text{SM}} \approx 7$ is an arithmetic consequence of the mirror branch values ($b'_2(\text{RHN}) = 29$ vs. SM $b_2(\text{RHN}) = 11$), not a tunable parameter. Once the structural gap theorem is proved, upgrading R_{dark} to a full structural prediction, this compressed hierarchy becomes a potentially falsifiable prediction for dark sector cosmology. Access to three dark neutrino mass-squared splittings with the ratio $R_{\text{dark}}/R_{\text{SM}} \approx 7$ would be strong evidence for the UGP mirror branch.

The dark neutrinos are sterile with respect to all SM gauge interactions. Their hierarchy is in principle accessible via dark sector precision cosmology (contributions to N_{eff} or the dark radiation spectrum) or via future experiments probing dark radiation at sufficient energy resolution.

5 Experimental Predictions and Detection Prospects

5.1 Prediction Table

Table 6 summarizes all experimentally accessible dark states predicted by the UGP mirror branch.

Table 6: UGP dark sector predictions. All masses are parameter-free; none are fitted. “Lean-cert” denotes machine-certification in Lean 4 (zero sorry); “GTE arith” denotes derivation from GTE arithmetic (not Lean-certified at the mass level); “cond.” denotes conditional on the structural gap theorem. $Q_{\text{EM}} = 0$ for all dark states is Lean-certified (zero axioms).

State	Mass	Q_{EM}	Cert. status	Primary search venue
GTE-P7	211.9 MeV	0	Lean-cert (P02)	BaBar/LHCb archival dimuon; Belle II Tier 1
Dark lepton G1 (χ_1)	0.5406 MeV	0	GTE arith	Sub-MeV DM; CMB spectral distortion (Higgs portal; below current sensitivity)
Dark lepton G2 (χ_2)	24.47 MeV	0	GTE arith	MeV-scale Higgs-portal DM; future dedicated search (below current sensitivity)
Dark lepton G3 (χ_3)	3604.68 MeV	0	GTE arith	LHC Higgs invisible width / mono-jet (λ_s suppressed; below current sensitivity)
Dark photon A'	Does not exist	—	Lean-cert	NA64/BaBar/Belle II null = consistent
$G2/G1$ mass ratio	45.4	—	GTE arith	Same experiment covering 0.5–25 MeV
R_{dark} (dark ν)	0.2080	—	Cond. on gap theorem	Dark radiation precision cosmology
Λ_{dark}	$\sim 200 \text{ MeV} - 1.7 \text{ GeV}$	—	α_s Lean-cert; N_f unknown	Dark hadron searches

5.2 Dark Hadron Subsector and the Confinement Scale

The dark $\text{SU}(3)$ coupling constant is *not* a free parameter. It is determined by the P01 Gauge Coupling Master Formula [8]:

$$g_G^2 = L_G \cdot D_G / 5^{\gamma_G}, \quad (8)$$

where L_G is the Weyl group order (fixed by group theory), D_G is the Vandermonde determinant squared (Elegant Kernel global MDL constant, not branch-specific), and $\gamma_G = 3$. For $\text{SU}(3)_{\text{dark}}$, the same inputs apply as for SM $\text{SU}(3)_c$:

$$\alpha_{s,\text{dark}}^{\text{bare}} = \frac{g_{3,\text{dark}}^2}{4\pi} = \frac{41,075,281}{27,648,000 \times 4\pi} = 0.11822 \quad (\text{same as SM}). \quad (9)$$

This structural equality— $\alpha_{s,\text{dark}}^{\text{bare}} = \alpha_{s,\text{SM}}^{\text{bare}}$ —follows from the shared Elegant Kernel and is not an assumption; it is derived from the same Vandermonde determinant that fixes the SM coupling. The formula is Lean-certified via `g3Sq_bare_eq` (zero sorry).

Given equal bare couplings, the dark confinement scale depends on the number of active dark quark flavors $N_{f,\text{dark}}$ at the running scale:

$$N_{f,\text{dark}} = 6 \Rightarrow \Lambda_{\text{dark}} \approx 210 \text{ MeV}, \quad (10)$$

$$N_{f,\text{dark}} = 2 \Rightarrow \Lambda_{\text{dark}} \approx 1.7 \text{ GeV}. \quad (11)$$

The value of $N_{f,\text{dark}}$ at the confinement scale is not yet determined from GTE arithmetic (dark G2/G3 quark masses are unknown); this introduces a range of [210 MeV, 1.7 GeV] for Λ_{dark} .

Under the $N_{f,\text{dark}} = 6$ scenario, dark pion masses (PCAC estimate) are $\approx 11 \text{ MeV}$ (up-type) and $\approx 59 \text{ MeV}$ (down-type), and the dark proton mass is $\approx 630 \text{ MeV}$ ($\approx 3 \times \Lambda_{\text{dark}}$). These are preliminary order-of-magnitude estimates.

The GTE-P7 state at 211.9 MeV [11] falls within the dark baryon mass range under the $N_{f,\text{dark}} = 6$ scenario, providing an internal cross-check: GTE-P7 is consistent with being a dark baryon or dark hadron excited state above Λ_{dark} .

5.3 Current Experimental Status and the Dark Photon Non-Prediction

No dark state from the UGP mirror branch is currently excluded by experiment. The reason is the predicted Higgs portal coupling:

$$\lambda_s \sim 10^{-6} \quad (\text{two-loop suppressed}), \quad (12)$$

which places all signal cross-sections 5–14 orders of magnitude below the current experimental sensitivity of BaBar, ATLAS/CMS, and the best sub-GeV dark matter searches.

The UGP dark sector is not in thermal equilibrium with the SM at any epoch. The FIMP production mechanism [4, 6]—freeze-in rather than freeze-out—means the dark sector never thermalized. It therefore does not contribute to the effective number of relativistic species N_{eff} in a way that would violate Planck 2018 constraints [1]. BBN constraints do not apply to the dark leptons ($Q = 0$, not in the SM thermal bath).

Dark photon non-prediction. A key structural prediction from the Mirror Braid Atlas is the *absence* of a dark photon. Since all mirror-branch particles have $Y_{\text{mirror}} = T_{3,\text{mirror}} = 0$ (Lean-certified in `DarkBraidAtlas.lean` and `DarkQuarkCharge.lean`), no dark $U(1)_D$ gauge boson exists, and kinetic mixing $\varepsilon = 0$ exactly by structure. Dark photon searches (NA64, BaBar, Belle II, FASER) are testing a class of models that the UGP dark sector predicts does not apply. A discovery of a dark photon signal would *falsify* the UGP dark sector structure at the highest confidence level. Continued null results are fully consistent with this prediction but do not constitute positive evidence for UGP until the two-peak signature (Section 5.4) is observed. The relevant UGP portal is the Higgs portal only.

5.4 The Two-Peak Signature

Dark singlet leptons G1 (0.54 MeV) and G2 (24.5 MeV) both fall within the 0.1–100 MeV experimental decade. An experiment covering this range with sufficient sensitivity would observe two signals simultaneously. The distinguishing feature is the fixed mass ratio:

$$\frac{m_{G2}}{m_{G1}} = \frac{24.47}{0.5406} \approx 45.3 \quad (\text{fixed, parameter-free}). \quad (13)$$

Any model can predict a single dark state. Predicting two specific states with a fixed ratio in the same experimental window is a much stronger falsification target. If both peaks are found but at

a ratio other than ≈ 45 , the UGP mirror branch arithmetic is falsified. The SM charged lepton analogue ($m_\mu/m_e \approx 207$) is compressed by a factor of ≈ 4.6 in the dark sector—a direct signature of the mirror branch b -value structure.

The GTE-P7 state at 211.9 MeV is the near-term priority target. The signal is a narrow resonance in decay channels accessible at BaBar, LHCb, and Belle II, with existing archival data not yet searched specifically for this state. A dedicated analysis is feasible without new data collection.

5.5 Relic Density: FIMP Overproduction and the Asymmetric Dark Matter Mechanism

The relic density question is the main open problem for P29’s cosmological falsifiability. We summarize the current status.

FIMP overproduction. A corrected semi-analytic calculation of the freeze-in yield for dark lepton G1 at $\lambda_s \sim 10^{-6}$ gives:

$$\Omega_{\chi_1} h^2 \approx 4.84 \times 10^6 \times (\Omega_{\text{DM}} h^2)_{\text{obs}}, \quad \lambda_s \sim 10^{-6}. \quad (14)$$

The FIMP freeze-in mechanism with the predicted portal coupling overproduces dark matter by approximately five million times the observed value $\Omega_{\text{DM}} h^2 = 0.120 \pm 0.001$ [1]. This result is robust to order-of-magnitude uncertainties in the parametric formula; a full Boltzmann calculation is required for a definitive result but would not change the overproduction by orders of magnitude.

Leading resolution: asymmetric dark matter. The most structurally motivated resolution is an asymmetric dark matter (ADM) mechanism [5], in which the dark matter abundance is set not by the FIMP yield but by a dark baryon asymmetry:

$$\eta_\chi \approx \frac{2}{7} \times \eta_{B+L, \text{pre-sphaleron}}. \quad (15)$$

The factor $2/7$ has a structural grounding in the UGP framework: $N_c \equiv 3 \pmod{7}$ (from the CUP-11a arithmetic of the dark sector’s mod-7 structure [10]), giving a dark baryon $\mathbb{Z}_7$ charge of $3 \times 3 = 9 \equiv 2 \pmod{7}$. The ratio $(\mathbb{Z}_7 \text{ charge of dark baryon})/(\mathbb{Z}_7 \text{ group order}) = 2/7$ provides a structural argument for the asymmetry factor.

GTE arithmetic baryogenesis rate. The pre-sphaleron baryon asymmetry can be estimated from GTE arithmetic. The three generation ridges at $n = 10, 13, 16$ (spaced by $N_c = 3$; Lean-certified in `GTBGenerationPrimes.lean`) each contribute a prime-selection probability $P_{\text{gen},i} \sim 1/\ln(c_{1,i})$. The c_1 seeds at all three ridges are prime (Lean-certified in `GTBGenerationPrimes.lean`; the six values are 823, 2137, 9007, 27817, 46681, 2489143). The resulting estimate is:

$$\eta_{B+L, \text{pre}} = N_f \times \prod_{i=1}^3 P_{\text{gen},i} \approx 3 \times 1.317 \times 10^{-6} \approx 3.95 \times 10^{-6}, \quad (16)$$

giving $\eta_\chi = (2/7) \times 3.95 \times 10^{-6} \approx 1.13 \times 10^{-6}$.

The required dark matter asymmetry (from the ADM formula $\Omega_\chi h^2 / \Omega_B h^2 = (m_\chi/m_p) \times (\eta_\chi/\eta_B)$) with $m_\chi = 0.54 \text{ MeV}$ is $\eta_\chi^{\text{req}} \approx 8.1 \times 10^{-7}$. The GTE estimate exceeds this by a factor of $1.39 \times$ (equivalently, e^{1/N_c} to 0.3%, a striking numerical coincidence without a current derivation). The estimate is within the O(factor 2) theoretical uncertainty of baryogenesis calculations—SM baryogenesis has uncertainties of 5–10 $\times$ from first principles.

Assessment. The GTE arithmetic formula Eq. (16) is a parameter-free order-of-magnitude estimate. It correctly predicts the size scale of the baryon asymmetry from pure arithmetic. The Lean-certified three-generation prime-lock structure forces the GTE winding numbers into a size window $[\mathcal{O}(10^3), \mathcal{O}(10^7)]$ that automatically produces the right scale. A derivation from first principles—analogue to the SM sphaleron calculation—remains the primary open question for the dark sector cosmology program. The full first-principles derivation of η_{B+L} from the GTE topological action, and a complete Boltzmann calculation of $\Omega_\chi h^2$, are deferred to future work (Sakharov conditions and topological baryon number violation in the GTE framework are discussed in [7]).

6 Conclusions

We have derived the Mirror Branch Braid Atlas—the complete particle classification for the UGP dark sector—and established the following results.

Machine-certified structural results.

1. $Q = 0$ for all dark particles, derived from Braid topology (not assumed). Lean 4: `MirrorWindingNumber.lean`, `DarkBraidAtlas.lean`, `DarkQuarkCharge.lean`.
2. Dark sector gauge group = $SU(3)_{\text{dark}}$ only. No dark electroweak symmetry. No dark photon ($\varepsilon = 0$ exactly by structure).
3. $c(W) = b_2(\text{RHN}) = q_1$ in both branches: first cross-sector structural unification in UGP Physics. Lean 4: `EWBosonRHNConnection.lean`.
4. Generation ridges at $n = 10, 13, 16$ spaced by $N_c = 3$, with the fourth slot $n = 19$ absent (consistent with the $N_f = 3$ ceiling from anomaly cancellation). Lean 4: `GTBGenerationPrimes.lean`.

Computationally derived predictions.

5. Dark lepton masses: 0.5406 MeV, 24.47 MeV, 3604.68 MeV (parameter-free; $Q = 0$ Lean-certified).
6. Dark lepton mass ratio $G_2/G_1 = 45.3$: compressed $4.6\times$ relative to SM $\mu/e = 207$; falsifiable two-peak signature.
7. Dark confinement scale $\Lambda_{\text{dark}} \in [210 \text{ MeV}, 1.7 \text{ GeV}]$ ($\alpha_{s,\text{dark}} = \alpha_{s,\text{SM}}$ Lean-certified; $N_{f,\text{dark}}$ pending).
8. FIMP relic density overproduction: $4.84 \times 10^6 \times$ at $\lambda_s \sim 10^{-6}$ (corrected semi-analytic).
9. Asymmetric dark matter formula $\eta_\chi = (2/7)\eta_{B+L,\text{pre}}$: GTE arithmetic gives $\eta_{B+L,\text{pre}} \approx 3.95 \times 10^{-6}$ (within $\mathcal{O}(\text{factor } 2)$ of the observed baryon asymmetry).

Conditional predictions (gap theorem pending).

10. Dark neutrino mass ratio $R_{\text{dark}} = 0.208$ — $7\times$ larger than SM $R_{\text{SM}} = 0.0294$ (conditional on structural derivation of $b'_3 = 37$).

Open problems for future work.

- Structural gap theorem: derive $b'_3 = q'_2 - b_{1,\text{RHN}}$ from GTE cascade rules, upgrading the dark neutrino hierarchy to a full structural prediction.
- GTE topological baryogenesis rate from first principles (GTE Lagrangian, instanton/ sphaleron calculation): needed to derive η_{B+L} analytically.
- Dark $SU(3)$ coupling running and confinement scale: determine $N_{f,\text{dark}}$ to pin down Λ_{dark} .

- Full dark hadron spectrum (dark pion, dark proton masses) following from Λ_{dark} .
- Full Boltzmann calculation of the dark lepton relic density.

Closing remark. The UGP dark sector is not a model but a prediction: the arithmetic sieve that selects the Standard Model necessarily produces this dark sector with the same number of free parameters—zero. The predictions are fixed: particle masses, quantum numbers, and the absence of a dark photon are all structural outputs of the GTE prime-lock, not free parameters. The most accessible near-term prediction is GTE-P7 at 211.9 MeV, a Tier 1 target at Belle II. The dark lepton sector (0.54, 24.5, 3600 MeV) has suppressed portal coupling ($\lambda_s \sim 10^{-6}$) placing it beyond current experimental reach, but the mass predictions and the fixed mass ratio ($G2/G1 \approx 45$) constitute sharp falsification targets as sensitivity improves. If the dark sector exists, no parameter can be adjusted to shift its properties; if it does not, no parameter can be adjusted to rescue the framework.

Appendices

A Lean Certificate Inventory

Table 7 provides the full inventory of Lean 4 certificates underpinning P29’s structural claims. All modules are machine-certified in the UGP Lean 4 library [14] with zero sorry.

Table 7: Full Lean 4 certificate inventory for P29. “Location” gives the file path within the UgpLean/ module tree of the UGP Lean repository. All entries have zero sorry in the compiled build. “Zero axioms” means no custom axioms beyond Lean/Mathlib foundations.

File	Key theorem(s)	Location	Axioms
MirrorWindingNumber.lean	mirror_lepton_winding_zero; mirror_lepton_q_em_zero (all gens); mirror_c1_ne_lepton_c1	BraidAtlas/	Zero axioms
DarkQuarkCharge.lean	dark_quark_charge_zero; dark_quark_charge_zero_all_gens; composite multi-generation certificate	BraidAtlas/	Zero axioms
DarkBraidAtlas.lean	$b'_1 = 5, b'_2 = 29, q'_1(\text{mirror}) = 29$; $SU(2)_L$ singlet structure; dark_ew_rhn_connection	BraidAtlas/	Zero sorry (definitions from P17 canonical modules)
EWBosonRHNConnection.lean	ew_rhn_connection_both_branches; ew_rhn_structural_connection; branch_shift_preserved	BraidAtlas/	Zero axioms
RHNGapTheorem.lean	$19 = 11 + 8, 37 = 29 + 8$ (arithmetic certificate; structural derivation open)	GTE/	Zero sorry
GTBGenerationPrimes.lean	12 theorems; all 6 c_1 values prime; gtb_generation_prime_certificate	GTE/	Zero sorry
P17 modules (ugp-lean)	prime_lock_unique_survivor; anomaly cancellation; SM charge derivation	ugp-lean (canonical)	Zero sorry

Note on the mirror winding number proof. MirrorWindingNumber.lean derives $Q = 0$ as a theorem from the GTE arithmetic definitions, with zero custom axioms. The $Q = 0$ result is a

consequence of the winding number formula $W_g = N_c \times (T_3 + Y/2) = 0$ for mirror-branch particles, proved by definitional unfolding; no physics axiom beyond the GTE integer arithmetic is needed.

B Dark Sector Mass Formula

The dark lepton masses in Table 3 are computed using the `InformationMassTransformer` from the UGP discovery engine, applied to the mirror branch lepton type with GTE cascade inputs from the mirror seed (1, 73, 2137). The formula is described in P01 [8] and P02 [11]; the mass formula itself is Lean-certified at the structural level in earlier papers. The numerical mass values are the arithmetic outputs of this formula, not independently Lean-certified. The SM sanity check passes to $< 0.01\%$ relative to PDG 2022 [15] values: $m_e = 0.511$ MeV, $m_\mu = 105.66$ MeV, $m_\tau = 1776.76$ MeV.

AI Disclosure

AI assistants were used for adversarial review, coding assistance, and debugging during manuscript preparation and code development. All scientific claims, derivations, and numerical results were independently verified by deterministic scripts and Lean proofs; no AI-generated content appears as unverified scientific output.

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Author Contributions

N.S. conceived the framework, performed all derivations, wrote the verifier code, and wrote the manuscript.

Competing Interests

The author declares no competing interests.

Data Availability

All data generated and analyzed during this study are included in this article and its supplementary information.

Code Availability

The computational infrastructure for this paper has three components.

Dark sector cascade. The mirror-seed mass cascade (dark lepton spectrum computation, SM sanity check) is implemented in:

```
papers/29_dark_sector_braid_atlas/artifacts/mdb_cascade.py
```

Relic density and FIMP yield. Semi-analytic freeze-in yield and relic density estimates are in:


```
papers/29_dark_sector_braid_atlas/artifacts/fimp_yield_analytic.py
papers/29_dark_sector_braid_atlas/artifacts/relic_density.py
```

Machine-checked proofs. The `ugp-physics-lean` Lean 4 library [14] formally verifies all structural claims, with zero `sorry` and zero custom axioms. The relevant modules are listed in Appendix A. Each theorem can be verified by cloning the repository and running `lake build`.

All components are versioned in public GitHub:

```
https://github.com/novaspivack/ugp-physics
https://github.com/novaspivack/ugp-physics-lean
```

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